Context is the New Weight: Distilling In-Context Adaptation into Weight Updates and Comparing the Mechanism at the Attention Layer

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Abstract

There are three obvious ways to make a language model behave according to a context C: prepend C at inference time (**in-context**); fine-tune the model to imitate the outputs it produces under C, with no C at inference (**context-distillation-FT**); or fine-tune the model on the text of C itself with a standard language-modelling loss (**context-FT**). The three interventions look superficially similar but produce very different changes in the underlying model. On Llama-3.1-8B-Instruct and five context types — structure constraint (high and low), style constraint, few-shot, and factual injection — we measure the resulting residual-stream perturbations site by site and find: context-distillation-FT recovers the in-context target both *behaviorally* (44–90% exact-match) and *mechanistically* (peak per-site cosine 0.44–0.79 in late layers, controlled against the base flow); context-FT moves more weight in absolute Frobenius norm but reproduces neither the behavior (0–13% EM) nor the mechanism (peak cosine 0.16–0.38). A no-behavior-change control further shows that the layer-1 weight-spike pattern observed in both fine-tunes is largely an optimizer-architecture artifact, not a signature of context internalisation. The three-setting comparison operationalises the in-context-learning-as-gradient-descent hypothesis on full instruction-tuned transformers, and shows that the equivalence holds for one specific recipe and breaks for the other.

1 Introduction

A language model can be adapted to behave according to a context C in several ways. The cheapest is to prepend C at inference time and let attention do the work — this is what we call **in-context** adaptation, or **Setting 1**. The most expensive is to fine-tune the model's weights so that the same behavior is baked in. But "fine-tuning to absorb C" is not a single recipe: at minimum, one can fine-tune on synthetic (Q, A) pairs in which A has been generated under C (**Setting 2: context-distillation-FT**), or one can fine-tune on the text of C itself with a standard language-modelling loss (**Setting 3: context-FT**). Settings 2 and 3 produce different θ_C, different ΔW, different residual-stream behavior. Both "count" as fine-tunes; both nominally "incorporate the context." Which one, if either, ends up at the same place as the in-context target?

This question is the empirical sibling of the in-context-learning-as- gradient-descent line of work [von Oswald et al., 2023, Akyürek et al., 2023, Garg et al., 2022]. Those papers establish, in linear-attention or other restricted regimes, that the in-context forward pass is mathematically equivalent to one step of gradient descent on a meta-objective. They do not perform the explicit gradient descent on the real transformer's weights and check whether the result matches; they characterise what Setting 1

Preprint.

is doing internally. We ask the explicit version: *do Settings 2 or 3, run as actual gradient-descent fine-tunes on the real model, end up at the same place in residual-stream space as Setting 1?*

For a fixed base model M (Llama-3.1-8B-Instruct) and five context types — structure constraint (high), structure constraint (low), style constraint, few-shot, factual injection — we run all three settings, then for each fine-tuned student we measure two perturbation vectors at every post-attention and post-MLP residual-stream site:

$$v_s^{\text{ctx}} = \text{resid}_s\big(Q' \mid \theta,\ [C; Q']\big) - \text{resid}_s\big(Q' \mid \theta,\ Q'\big),$$
$$v_s^{\text{dw}} = \text{resid}_s\big(Q' \mid \theta_C,\ Q'\big) - \text{resid}_s\big(Q' \mid \theta,\ Q'\big).$$

The first is what context did to the residual stream (Setting 1); the second is what ΔW did (Setting 2 or Setting 3). Both vectors live in the same basis, so the cosine between them is a direct measure of mechanism alignment.

Headline. Across all five contexts and all 64 sites of the 32-layer model: context-distillation-FT (Setting 2) reaches peak per-site RMSNorm cosine 0.44–0.79 in late layers (24–31), with 44–90% exact-match behavioral reproduction on training queries. Context-FT (Setting 3) reaches peak cosine 0.16–0.38 and 0–13% EM, despite producing a 4–$8\times$ larger Frobenius weight delta. A no-behavior-change control shows that the layer-1 concentration in both fine-tunes' ΔW is mostly an optimizer-architecture artifact rather than a signature of context internalisation. Setting 2 is the only schema among the three fine-tunes that approximates Setting 1 both behaviorally and at the mechanism level.

Contributions.

1. **Three-setting framework** for context internalisation (in-context, context-distillation-FT, context-FT) that allows distinguishing "training to imitate context-induced outputs" from "training on the context text."
2. **Behavioral and mechanism-level comparison** of the three settings, on a real instruction-tuned 8B transformer rather than a toy linear-attention model. Establishes that the in-context-learning-as-gradient-descent equivalence holds empirically for one specific fine-tune recipe (Setting 2) but breaks for naive language-modelling on the context (Setting 3).
3. **Site-by-site map** of where context lives in Llama-3.1-8B, measured by the alignment between v^{ctx} and v^{dw} at every post-attention and post-MLP residual-stream site, complemented by a triangle reading that includes the base-flow component.
4. **Optimizer-artifact control**: distillation on the base model's own no-context outputs, ruling out the trivial "any fine-tune lands at layer 1" alternative for the ΔW-magnitude pattern.
5. **Behavior-verification gate** that requires a fine-tuned student to reproduce the in-context target on its own training set before any mechanism analysis runs.

2 Setup

Model. Llama-3.1-8B-Instruct [Dubey et al., 2024], used in bf16 precision. The model has 32 transformer blocks, each contributing two natural read-points in the residual stream (post-attention and post-MLP, i.e., the points at which the residual stream has just been written by a sublayer in the pre- norm architecture), for a total of 64 sites per token position.

Contexts. We pick five contexts to span a deliberate range:

- **Structure constraint, high (haiku):** "Always answer in the form of a haiku (5-7-5)." Imposes a rigid syllabic and line structure on every output.
- **Structure constraint, low (concise):** "Be extremely concise. One short sentence, no preamble." A weaker, length-only structural constraint.
- **Style constraint (pirate):** "You are a pirate. Speak only in pirate slang." Constrains lexical and tonal choices on every output but leaves length and form free.
- **Few-shot (translate-fr):** three worked English-to-French examples.

- **Factual injection:** a system prompt that asserts five fictional facts (e.g., a password, a city of residence).

Three model-change schemas. We compare three ways to make the model behave according to C:

- **Setting 1 (in-context):** run base M with $[C; Q]$ at inference. No training. The behavioral target.
- **Setting 2 (context-distillation-FT):** fine-tune M on synthetic (Q, A) pairs where $A = M([C; Q])$. The student sees Q alone and is trained to reproduce A. The student does not see C at inference.
- **Setting 3 (context-FT):** fine-tune M with standard language- modelling loss on the chat-template encoding of C itself, no (Q, A) pairs. The student again does not see C at inference.

Sections 2–4.2 focus on Settings 1 and 2; Section 4.3 reports Setting 3 as a baseline that isolates the contribution of the synthetic (Q, A) supervision.

Synthetic dataset. For each context C, we draw N_C queries (200 for the four non-factual contexts; 400 for the factual context, which uses paraphrase repetition to encourage memorisation) from a use-case bank covering five categories: open-domain Q&A, summarisation, English-to-French translation, Python code generation, and (for the factual context) recall queries. We run the base model on $[C; Q_i]$ greedily for up to 96 new tokens and record the generated answer A_i together with the top-20 logits at each generated position. Refusals and degenerate outputs (length < 3 tokens) are filtered.

Distillation (Setting 2 recipe). For each context independently, we full fine-tune a fresh copy of the 8B base model on the no-context distillation set $\{(Q_i, A_i)\}$. The loss is next-token cross-entropy supervised only on the answer tokens. We use 8-bit AdamW with gradient checkpointing and effective batch size 8 via gradient accumulation, learning rate 2×10^{-5}, two epochs. The Setting 3 (context-FT) recipe shares all hyperparameters but trains on the chat-template encoding of C alone for 100 forward-backward passes (see Section 4.3).

Behavior verification (training set). Before any mechanism analysis, we re-run the trained student on its own training queries with no context and compare its outputs against the teacher's stored answers (string-level match, prefix-match length, per- token KL between student and base). Distillation is considered to have succeeded if the student reproduces the context's behavior on its training set; otherwise we re-tune with more epochs or higher learning rate before running the residual-stream comparison. Concretely, we require an exact-match rate of at least 0.30 or a prefix-5 match rate of at least 0.50 on the training queries before proceeding to Section 4; in our runs all five contexts cleared this threshold (Table 2).

Held-out validation. Training-set verification could be memorisation. We therefore evaluate on a separate validation set of 30 queries per context that are drawn from five use cases (Q&A, summarisation, translation, code, recall) and are *disjoint* from the training pool of Section 2. For each query Q' we generate three answers — the base $A_{\text{base}} = M(Q' \mid \theta, \emptyset)$, the in-context teacher $A_{\text{ctx}} = M(Q' \mid \theta, [C; Q'])$, and the context-distillation-FT student $A_{\text{stu}} = M(Q' \mid \theta_C, \emptyset)$ — and report two complementary signals.

The first is the ROUGE-L F1 between pairs and the resulting *ROUGE gap-closure*: $(\mathrm{R}_L(A_{\text{stu}}, A_{\text{ctx}}) - \mathrm{R}_L(A_{\text{base}}, A_{\text{ctx}}))/(1 - \mathrm{R}_L(A_{\text{base}}, A_{\text{ctx}}))$. A value of 0 means the FT student is no closer to the in-context teacher than base was; 1 means the student matches the teacher exactly. The second is a *behavioural pattern rate* appropriate for the context type — fraction of three-line outputs for haiku, presence of pirate vocabulary for pirate, output length cap for concise, French-character or French-word presence for translate-fr, refusal-pattern rate for factual — and the analogous behavioural gap-closure $(p_{\text{stu}} - p_{\text{base}})/(p_{\text{ctx}} - p_{\text{base}})$.

Table 1 shows that for the four non-factual contexts the student reproduces the teacher's behavioural pattern on held-out queries at a rate well above base ($+0.77$ to $+1.00$ behavioural closure), confirming that the training-set EM in Table 2 is not pure memorisation. The factual context is a special case: the

Table 1: Held-out validation. ROUGE-L gap-closure measures how much of the base→teacher distance the student covers in surface tokens. Behavioural pattern rate measures whether the structural feature characteristic of the context type (haiku-shape, pirate vocabulary, short answer, French content, refusal) appears at the rate the teacher exhibits. Behavioural gap-closure is the analogous fraction in pattern space. ROUGE-L underrates stylistic transfer where content drifts but form is preserved (haiku, fewshot); the behavioural rate captures this.

Context	R_L(base, ctx)	R_L(stu, ctx)	Pattern (T/S/B)	ROUGE-L closure	Beh. closure	status
haiku	0.226	0.523	0.93 / 0.75 / 0.04	0.384	+0.80	pass (beh.)
pirate	0.322	0.610	1.00 / 1.00 / 0.14	0.424	+1.00	pass
concise	0.421	0.661	0.96 / 0.96 / 0.29	0.414	+1.00	pass
fewshot_translate_fr	0.459	0.767	0.64 / 0.54 / 0.18	0.570	+0.77	pass
factual	0.364	0.515	0.44 / 0.22 / 0.52	0.238	ambiguous	marginal

teacher's own behaviour on out-of-distribution queries is inconsistent (sometimes refusing, sometimes answering with "I can provide general information", sometimes refusing translation requests on the grounds that the system prompt is about confidential facts), so the refusal-rate comparison does not give a clean target for the student to match. We retain factual in subsequent analysis but flag this as a caveat.

KV-vs-ΔW probe set. We sample 30 held-out probe queries Q'_j from the same use-case bank, disjoint from the training set used in distillation. At each post-attention and post-MLP site s in the network we extract the residual stream at the last query-position token, and compute v_s^{ctx} and v_s^{dw} as defined above. Comparison metrics:

- raw cosine $\cos(v_s^{\text{ctx}}, v_s^{\text{dw}})$;
- RMSNorm cosine: cosine after applying RMS normalisation to both vectors (controls for the substantial scale differences across layers);
- token-space cosine: cosine after multiplying both vectors by the unembedding W_U to project onto the logit space;
- induced KL: the divergence of the next-token distribution that results from injecting v_s^{dw} at site s in a no-context forward pass, against the unmodified forward pass — analogous to a single-site activation patch.

3 Behavioral equivalence: does Setting 2 reproduce Setting 1?

We start with the most basic question: does the context-distillation-FT student reproduce the in-context target's outputs? Section 4.3 extends the same comparison to context-FT.

For each context we run Setting 2's four-phase pipeline. Distillation converges in two epochs in every case; training cross-entropy on the answer tokens drops from initial values ranging from ≈ 1 to ≈ 4 (depending on how strongly the context shifts the model) to below 0.05 in all five experiments.

Table 2 reports the post-distillation behavior verification on each context's own training queries. The two structure-constraint contexts (haiku, concise) achieve near-perfect reproduction — 89–90% exact-match between the no-context student and the with-context teacher — confirming that the context-distillation-FT weights have internalised the behavior change. The few-shot (translate-fr) context follows at 0.795 EM. The factual context is harder (0.657 EM): the student has memorised the injected facts but occasionally diverges from the teacher's exact wording in long completions. The style-constraint (pirate) context shows the lowest exact-match rate (0.440) but a high prefix-match rate (0.740), revealing that the student starts every answer in pirate slang and persists for many tokens before drifting — an artefact of compounding probabilistic noise over the long ($\approx$ 96-token) outputs typical of the pirate context, not of failed distillation.

Table 2: Behavior verification on training queries. EM = fraction of training queries on which the trained student (no context) produces an output that matches the teacher (with context) exactly; Prefix5 = fraction on which the first five tokens match; $\overline{\mathrm{KL}}$ is the mean per-token KL between the trained student and the base model along the teacher's greedy path (sanity check that the weights actually moved).

Context	Type	N	EM	Prefix5	$\overline{\mathrm{KL}}$
haiku	structure (high)	200	0.895	0.970	3.21
pirate	style	200	0.440	0.740	1.30
concise	structure (low)	200	0.900	0.960	1.05
fewshot_translate_fr	few-shot	200	0.795	0.820	0.59
factual	factual	175	0.657	0.800	2.85

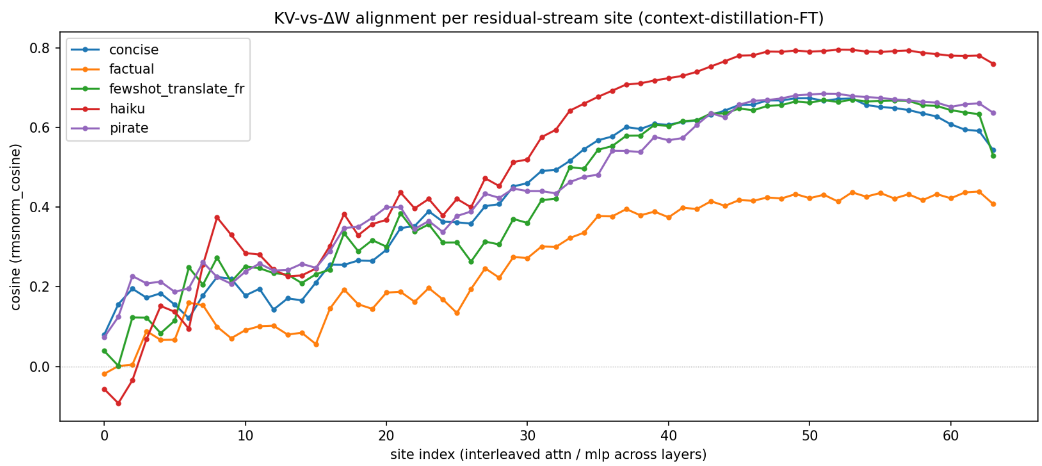

Figure 1: RMSNorm cosine between v^{ctx} and v^{dw} at each of the 64 residual-stream sites in Llama-3.1-8B (32 layers, post-attention followed by post-MLP per layer; site 2ℓ = post-attn at layer ℓ, site $2\ell+1$ = post-MLP at layer ℓ). Averaged over a held-out probe set of 28 queries per context. One line per context type.

4 Mechanism alignment: does Setting 2's ΔW act like Setting 1's KV?

Behavioral equivalence is necessary but not sufficient: a student can reproduce the teacher's outputs by an entirely different internal route. We now ask whether context-distillation-FT also reproduces the in-context *mechanism*: at the layers where prepending C shifts the residual stream, does the corresponding ΔW shift it the same way?

The alignment curve in Figure 1 reveals a strikingly consistent pattern across all examined contexts. Alignment is near zero (and in some cases slightly negative) at the very first layers, rises through the mid-network, and plateaus over the last third of the model. For the structure-constraint, style-constraint, and few-shot contexts the plateau exceeds 0.65 in RMSNorm cosine; for the factual context it is lower (around 0.4) but the peak sits even later in the network. This is consistent with the intuition that surface-form constraints are dictated by late-network "surface" computation, while factual recall recruits the very last attention layer just before the unembedding.

Table 3 reports the peak-alignment site per context. All peaks cluster in layers 24–31 of the 32-layer model. The two structure-constraint contexts (haiku, concise) peak at an attention site (L26_ATTN, L25_ATTN). The style-constraint (pirate) and few-shot contexts peak at MLP sites (L25_MLP, L26_MLP), suggesting that contexts that induce more elaborate representation-binding — maintaining a voice, or applying a demonstrated transformation to a new input — recruit the MLP rather than the attention sublayer. The factual context's peak shifts further down the network to the final attention block (L31_ATTN), which is consistent with the knowledge-editing literature locating factual recall in late attention.

Table 3: Peak-alignment site per context. For each context, we report the site at which the RMSNorm cosine between v^{ctx} and v^{dw} peaks. Llama-3.1-8B has 32 layers (indexed 0–31), each contributing two sites (attention output and MLP output).

Context	Peak layer	Peak sublayer	Peak cosine
haiku	26	attn	0.795
pirate	25	mlp	0.685
concise	25	attn	0.674
fewshot_translate_fr	26	mlp	0.670
factual	31	attn	0.439

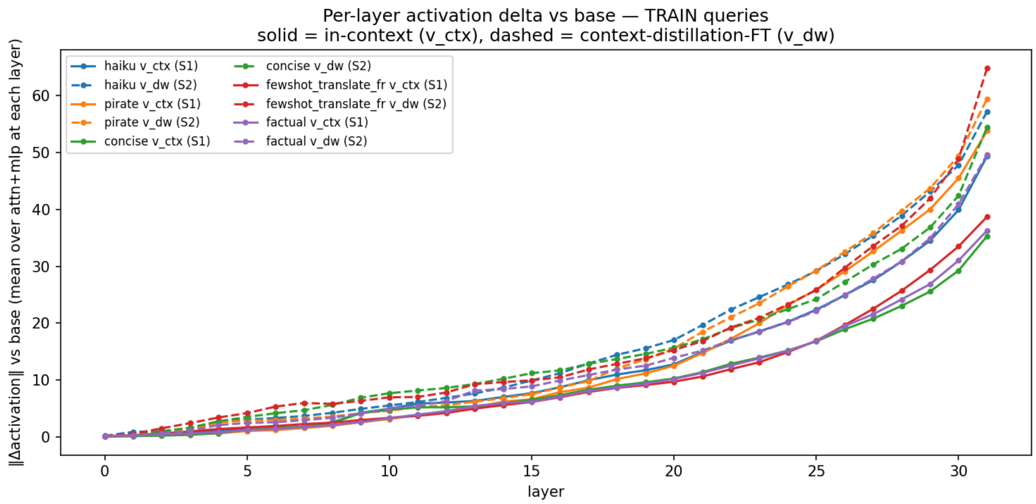

Figure 2: Per-layer activation delta magnitude vs base, on **training** queries. Solid = $\|v^{\text{ctx}}\|$ (in-context perturbation, S1). Dashed = $\|v^{\text{dw}}\|$ (context-distillation-FT perturbation, S2). The FT student perturbs the residual stream noticeably more than the in-context teacher does at every layer, with the gap growing in the last six layers.

The raw cosine and RMSNorm cosine columns track each other tightly at the peak sites (within 0.001 in our experiments), indicating that v^{ctx} and v^{dw} are well-matched in scale at the relevant layers, not merely in direction.

4.1 Activation-magnitude profile across layers

The cosine measurement leaves two questions open: how large are v^{ctx} and v^{dw} in absolute terms, and does the student's activation pattern depend on whether the query was seen during training? Figures 2–5 report per-layer L2 norms for both perturbation vectors, on TRAIN queries (sampled from each student's training set) and on disjoint VAL queries (from Section 3's held-out set), averaged over attn and MLP sites at each layer.

Three observations from these four figures:

S2 perturbs activations more than S1. At every layer, $\|v^{\text{dw}}\|$ exceeds $\|v^{\text{ctx}}\|$, with the gap widening in the last six layers (Figures 2 and 3). The context-distillation-FT student does not merely reproduce what context did; it overshoots in magnitude. Combined with the high cosine alignment in Figure 1, this means the two perturbations point in similar directions but the student's vector is longer.

Train and validation activations match. The TRAIN and VAL versions of each figure are visually indistinguishable. The FT student's residual-stream perturbation has the same magnitude profile on queries it was trained on as on held-out queries it has never seen. This rules out the trivial explanation that high alignment is an artefact of memorising the training distribution.

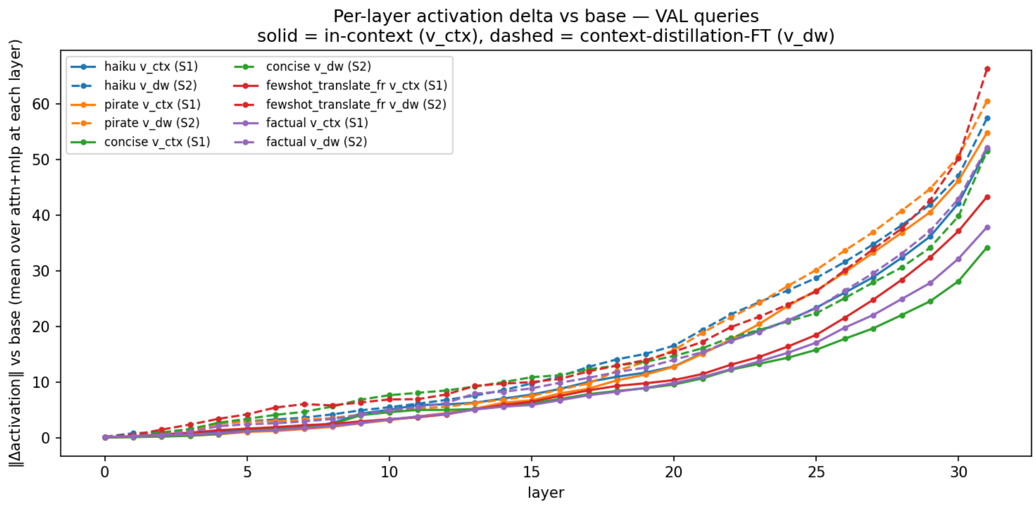

Figure 3: Same as Figure 2, on **held-out validation** queries. The curves are nearly identical to the training-set version, indicating that the FT student's activation pattern generalises rather than memorising training examples.

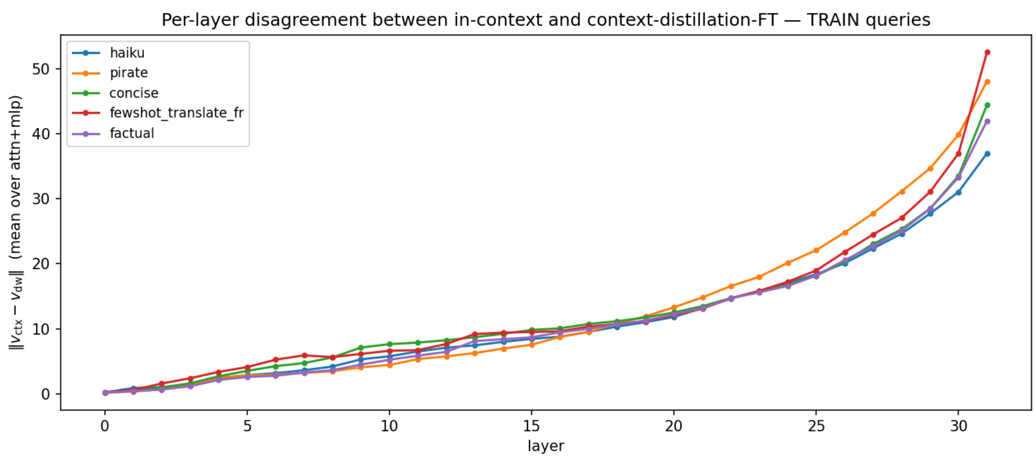

Figure 4: Per-layer disagreement $\|v^{\text{ctx}} - v^{\text{dw}}\|$ on **training** queries. Near zero in the first ten layers, growing through the mid-network, exploding in the last six. Pirate and few-shot peak at ≈ 50 at layer 31; haiku ends at ≈ 37.

Disagreement is concentrated in the last six layers. Figures 4 and 5 show that $\|v^{\text{ctx}} - v^{\text{dw}}\|$ is small (under 10) for layers 0–25 across all contexts, then grows sharply in layers 26–31. The same layers that are most behaviourally important also exhibit the largest divergence between the two interventions in absolute Euclidean terms. So while context-distillation-FT and in-context point in similar directions where it counts (high cosine), the magnitude of disagreement is also concentrated there.

4.2 Where the weights actually moved

The alignment curve tells us *where the weight delta has the same effect as context*, but not *where the weights changed*. To complete the picture we compute, per layer ℓ, the Frobenius norm $\|\theta_C^\ell - \theta_0^\ell\|_F$ of the weight delta restricted to that layer's parameters.

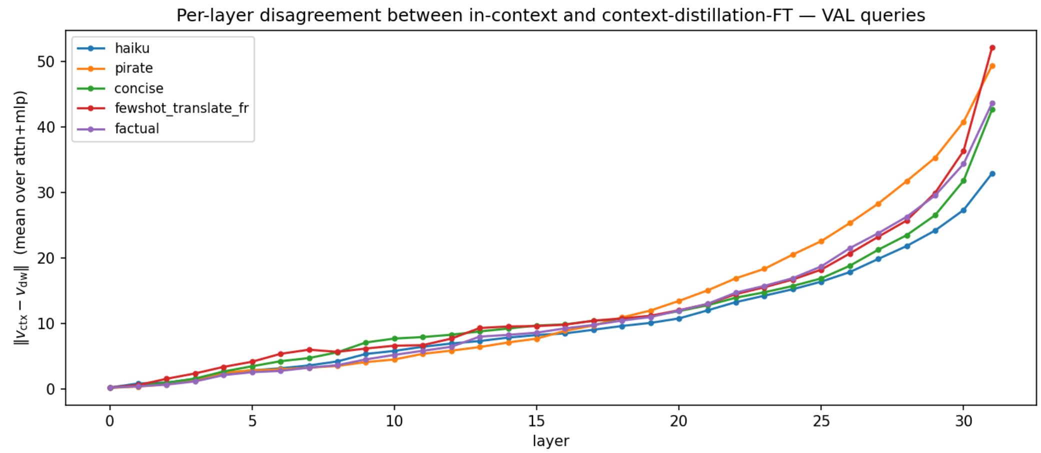

Figure 5: Same as Figure 4, on **held-out validation** queries. Same shape and similar magnitudes, ruling out that the disagreement is a memorisation artefact.

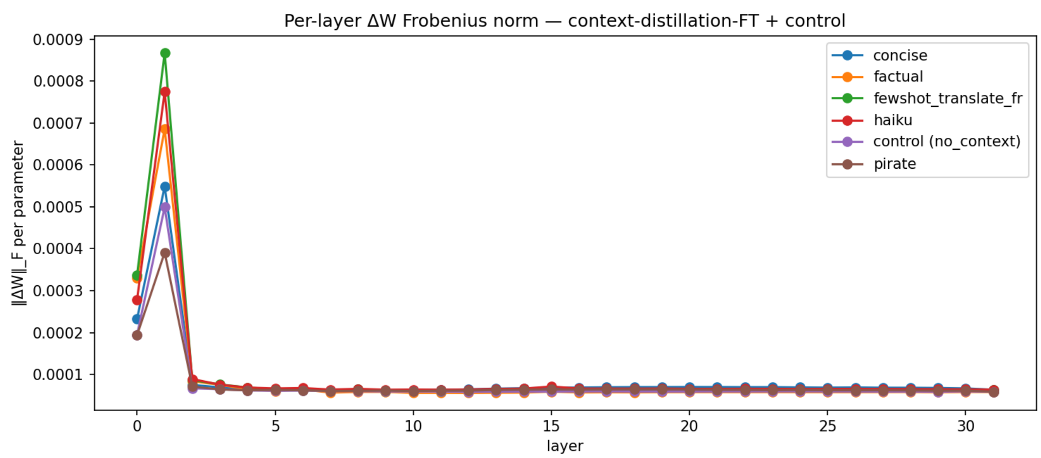

Figure 6: Per-parameter RMS of $\Delta W = \theta_C - \theta_0$ at each layer, for each of the five contexts. Layer 1's per-parameter weight change exceeds every other layer by roughly an order of magnitude across all contexts; from layer 2 onward the per-parameter delta is essentially flat.

Figure 6 shows a striking dissociation from Figure 1: **the weights move almost exclusively at layer 1**, with an order-of-magnitude per-parameter spike, while layers 2–31 show a near-uniform small delta. Yet the alignment between v^{ctx} and v^{dw} peaks not at layer 1 but at layers 24–31. The early-layer change in θ_C propagates forward through the residual stream, and its accumulated effect by late layers happens to match what the context's KV cache does at those layers.

A breakdown of ΔW by parameter group (attention projections, MLP projections, layer norms) finds that `mlp.down_proj` accounts for the largest fraction of the total Frobenius mass in every context, followed by `attn.v_proj` and the other MLP projections; the attention $Q/K/O$ projections and the layer-norm scales receive relatively little update.

Table 4 sharpens the finding numerically. Layer 1 alone holds 50–76% of the total squared Frobenius mass of ΔW in every context; layers 0–1 together hold 62–88%. Yet the cosine-aligned site sits 24–30 layers downstream.

Control: layer-1 concentration is mostly an optimizer artifact. The bottom row of Table 4 reports a fine-tune of the base model on its *own* no-context outputs — an identical training recipe applied to data the model already produces, with essentially no behavior to internalize (mean per-

Table 4: Where weights moved vs where alignment peaked. "Total $\|\Delta W\|_F$" is the Frobenius norm of $\theta_C - \theta_0$ over all 8.0 B parameters. "L1 share" is the fraction of $\|\Delta W\|_F^2$ contributed by layer 1 (square-of-Frob is additive across disjoint parameter groups). "(L0+L1) share" adds layer 0. The bottom row is the control: distillation on the base model's own no-context outputs — a fine-tune that has *no* behavior change to internalize.

Context	Total $\|\Delta W\|_F$	L1 share	(L0+L1) share	Peak alignment site	Hops to peak
haiku	13.53	73.9%	83.4%	L26_attn (0.795)	25
pirate	8.30	49.5%	61.6%	L25_mlp (0.685)	24
concise	10.42	61.3%	72.3%	L25_attn (0.674)	24
fewshot_translate_fr	14.81	76.1%	87.6%	L26_mlp (0.670)	25
factual	12.30	68.5%	84.3%	L31_attn (0.439)	30
control (no_context)	9.37	63.0%	72.6%	—	—

Table 5: Three model-change schemas, side by side. Setting 1 is the in-context target. Setting 2 is the context-distillation-FT student of Section 3. Setting 3 is fine-tuning on the context string only, with no (Q, A) pairs.

Context	Setting	EM	Peak alignment	Total $\|\Delta W\|_F$	L1 share
haiku	S2 (context-distillation-FT)	0.895	L26_attn (0.795)	13.53	73.9%
	S3 (context-FT)	0.025	L26_mlp (0.311)	55.60	82.1%
pirate	S2	0.440	L25_mlp (0.685)	8.30	49.5%
	S3	0.000	L10_mlp (0.318)	50.98	85.2%
concise	S2	0.900	L25_attn (0.674)	10.42	61.3%
	S3	0.135	L26_mlp (0.378)	89.08	86.4%
fewshot_translate_fr	S2	0.795	L26_mlp (0.670)	14.81	76.1%
	S3	0.000	L10_mlp (0.321)	52.48	88.6%
factual	S2	0.657	L31_attn (0.439)	12.30	68.5%
	S3	0.029	L03_attn (0.163)	38.43	85.3%

token KL between control student and base is 0.13, an order of magnitude smaller than for the haiku context's 3.21). Despite that, the control's layer-1 share (63%) and total Frobenius (9.37) sit *within the range* spanned by the five context fine-tunes. The layer-1 spike is therefore largely a property of the optimizer-architecture pair, not a signature of context internalisation. What context-induced fine-tuning adds is additional weight motion on top of this baseline pattern — 20–60% extra Frobenius mass for the structure- and style-constraint contexts — distributed similarly across layers but with the extra coverage that produces the late-layer alignment seen in Figure 1.

What is compact, then. The honest claim is not that context-internalising fine-tunes are compact; it is that *any* fine-tune of this model under this recipe lands disproportionately at layer 1. The contexts retain their distinguishing signature in the *alignment* measurement (Figure 1), where the control has no analogue, and in the magnitude of the additional weight motion above the no-op baseline.

4.3 Setting 3 (context-FT) — fine-tune on the context itself

Section 2's distillation recipe ("Setting 2") trains on (Q, A) pairs in which A has been generated under context. A natural question is whether we even need the (Q, A) side: can we just *train on the context string itself*, applying standard language-model loss to the chat-template-formatted system prompt (and few-shot demos, where present) without producing any synthetic dataset? We call this **Setting 3** and report it for all five contexts. Hyperparameters are matched to Setting 2 (lr 2×10^{-5}, 8-bit AdamW, gradient checkpointing, gradient accumulation 8); the only differences are that the training sequence is the chat-template encoding of C and that we run 100 forward-backward passes (limited by how quickly loss saturates on a single short sequence).

Three observations from Table 5 and Figure 7:

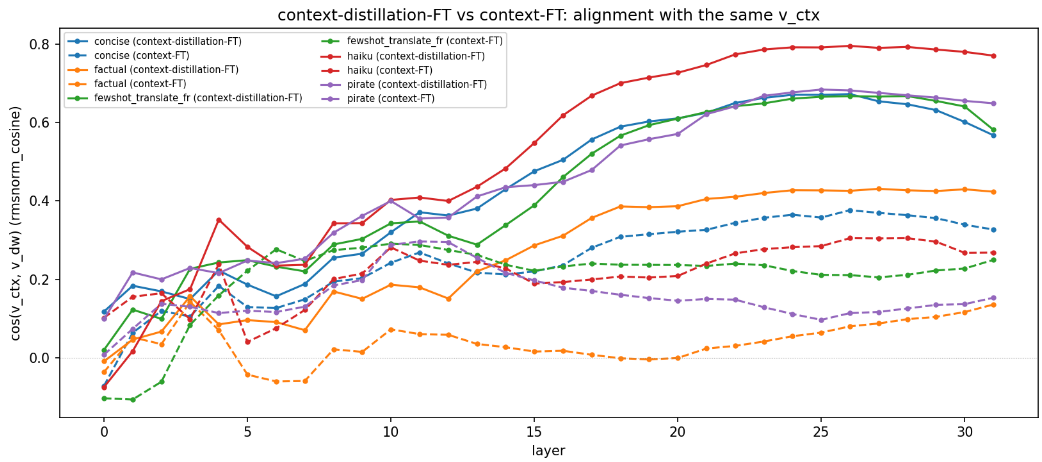

Figure 7: Per-layer cosine between v^{ctx} and v^{dw}, comparing Setting 2 (context-distillation-FT, solid) and Setting 3 (context-FT, dashed). Both are evaluated against the same v^{ctx} (i.e., the same context applied at inference). Setting 3 plateaus at 0.15–0.38 across contexts; Setting 2 reaches 0.55–0.79 at the same sites.

Setting 3 does not transfer the behavior. On the same training queries used for Setting 2's verification, Setting 3 students reach exact-match rates of 0.000–0.135 — effectively zero relative to Setting 2's 0.44–0.90. Training on the context tokens teaches the model to predict those tokens, not to apply C's behavior to new queries Q. The synthetic (Q, A) supervision is doing essential work.

Setting 3's v^{dw} aligns weakly but non-trivially with v^{ctx}. The peak alignment cosines drop from 0.44–0.79 (Setting 2) to 0.16–0.38 (Setting 3). The S3 alignment is well above zero, indicating that training on the context string does push the residual stream in *roughly* the right direction — just much less effectively. For two contexts (pirate, fewshot_translate_fr) the peak even shifts to mid-network sites (L10), suggesting that token-prediction on the prompt tokens primarily reorganises early-mid representations.

Setting 3 moves the weights more in absolute terms. Total $\|\Delta W\|_F$ for Setting 3 is 4–$8\times$ larger than for Setting 2. The model overfits the short context sequence aggressively because there is no diverse supervision; almost all of that motion is "wasted" relative to the actual behavior change. Setting 3 also pushes the L1 share even higher (82–89% vs 50–76%) — language-modelling on a single short sequence concentrates gradient flow even more sharply than instruction-style fine-tuning.

Summary. Of the three model-change schemas, only Setting 2 both reproduces the in-context behavior *and* aligns at the mechanism level. Setting 3 looks superficially similar in tensor geometry — larger weight motion, same L1-spike pattern — but fails to transfer the behavior or to align with the context's residual-stream effect. "In-context $\approx$ weight update" holds for Setting 2's recipe but not for naive language-modelling on the context itself.

5 Related work

In-context learning as implicit gradient descent. von Oswald et al. [2023], Akyürek et al. [2023], and Garg et al. [2022] establish, in linear- or restricted-attention settings, that a forward pass over an in-context demonstration is mathematically equivalent to one step of gradient descent on a meta-objective. They characterise what Setting 1 is doing internally, without ever performing the explicit gradient descent on the real transformer's weights. Our work runs the explicit version of the experiment: Settings 2 and 3 *actually* perform gradient descent on the model's weights with two different losses, and we measure how close each ends up to the in-context target in residual-stream space. The cosine 0.44–0.79 for Setting 2 supports the qualitative prediction of vO et al. on a real instruction-tuned transformer; the cosine 0.16–0.38 for Setting 3 shows the equivalence is recipe-specific rather than a property of any fine-tune.

Knowledge editing. ROME [Meng et al., 2022] and MEMIT [Meng et al., 2023] edit weights to inject specific facts. Our factual-context experiment is the dual operation: rather than choosing the edit, we let SGD discover one and ask whether the result resembles the context that prompted it.

Soft prompts and prompt distillation. Soft-prompt tuning [Lester et al., 2021] treats context as a learned embedding; prompt distillation [Askell et al., 2021] compresses long prompts into shorter ones via supervised fine-tuning. The latter is methodologically close to our distillation step, but it is consumed as a training procedure, not as a mechanistic probe.

6 Discussion

Which fine-tune recipe matters. The headline result of this paper is that Settings 2 and 3 are not interchangeable, despite both nominally being "a fine-tune that incorporates the context." Context-distillation-FT (Setting 2) reaches the in-context target both behaviorally and at the mechanism level; context-FT (Setting 3) reaches neither. Practitioners who intuitively treat "training on the prompt" as a substitute for "training on demonstrations of behavior under the prompt" should not. The difference is not just in output quality — it is in where the resulting weight delta lives in residual-stream space.

Implications for the in-context-learning-as-GD literature. The line of work from von Oswald et al. [2023] and follow-ups predicts that explicit GD on a suitable meta-loss should match Setting 1. Our Setting 2 cosine of 0.44–0.79 is consistent with that prediction extended to a real 8B transformer with a real loss; the gap from 1.0 suggests either that real CE-on-tokens is not exactly the meta-loss those papers identify, or that 32 nonlinear layers leave many descent directions equivalent for the same behavior. Setting 3 sits well below that range, ruling out the trivial "any fine-tune lands at the same place" alternative.

Implications for unlearning. If Setting 2's ΔW approximates context's residual-stream effect, *subtracting* a ΔW obtained from distilling a piece of unwanted context might be equivalent to redacting that context at inference. We do not test this directly, but the geometry suggests it is at least worth attempting.

Future work: prompt inversion. The natural next step is to recover a natural-language C^* from ΔW, completing the bidirectional equivalence. The triangle geometry we report ($\mathrm{v}^{\mathrm{ctx}}$ and v^{dw} both anti-aligned with the base flow at similar angles, similar magnitudes, and high mutual cosine) is the kind of structure soft-prompt optimisation should be able to exploit.

7 Limitations

- Single model family (Llama-3.1-8B-Instruct). Whether the three-setting comparison transfers to other architectures is open.
- Single training-set size, learning rate, and step count per setting. Setting 3's cosine could plausibly improve with more compute (longer training of context-FT, additional repetitions of the context text); we have not characterised how each setting's alignment scales.
- Setting 2's synthetic dataset is generated by the same base model that is being fine-tuned, so the distillation is closer to behavior cloning of self-outputs than to imitation of an external teacher; results may underestimate the difficulty of context internalisation in general.
- Setting 1 is the only schema for which v^{ctx} is defined; Settings 2 and 3 are compared via the v^{dw} each induces. We do not have a third orthogonal probe (e.g. a soft-prompt-recovered context for each ΔW) that would close the triangle.
- Single run per (context, setting) configuration. No error bars across seeds or training-recipe variations.
- Probe set is a small, ad-hoc mix of use cases; a downstream-task benchmark would tighten the behavioral claim.

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